

JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	SN1 and SN2 Nucleophilic Substitution	18 PYQs	Reaction Mechanisms
Standard	Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	1 PYQs	Elimination & Aromatic

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
SN1 and SN2 Nucleophilic Substitution	$\text{SN1: Rate} \propto [\text{RX}]$; $(3^\circ > 2^\circ > 1^\circ, \text{Racemization})$, $\text{SN2: Rate} \propto [\text{RX}][\text{Nu}]$; $(1^\circ > 2^\circ > 3^\circ, \text{Inversion})$
Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions	Saytzeff Rule: Highly substituted, more stable alkene is the major product in dehydrohalogenation

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 SN1 and SN2 Nucleophilic Substitution (Very High Yield — 18 PYQs)

Core Principle: Kinetics, stereochemical outcomes, carbocation rearrangements in SN1, and concerted backside displacement in SN2.

■ **Caution / JEE Pitfall:** Polar PROTIC solvents (e.g. H_2O , ROH) stabilize carbocations and favor SN1. Polar APROTIC solvents (e.g. acetone, DMSO) favor SN2 by desolvating nucleophiles.

■ **Speed Trick:** Allylic and benzylic halides undergo BOTH SN1 (resonance-stabilized carbocation) and SN2 (low steric hindrance) at remarkably high rates.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is..." [Ans: D]
- [2026 Shift] Q#2 (MCQ): "In $\text{S}_{\text{N}}1$ reaction, the alkyl halide that on hydrolysis produces racemic mixture is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "The compound from which chlorobenzene cannot be prepared easily is..." [Ans: N/A]

3.2 Elimination (E1/E2, Saytzeff vs Hofmann) & Haloarene Reactions (Standard Yield — 1 PYQs)

Core Principle: β -elimination mechanism, Saytzeff vs Hofmann regioselectivity, and unreactivity of haloarenes toward nucleophilic substitution.

■ **Caution / JEE Pitfall:** Haloarenes (like chlorobenzene) are extremely UNREACTIVE toward nucleophilic substitution due to partial double bond character (C-Cl resonance) and sp^2 carbon.

■ **Speed Trick:** Presence of electron-withdrawing groups (like $-NO_2$) at ortho- and para-positions dramatically increases reactivity of haloarenes toward nucleophilic substitution.

Representative Source PYQs:

- [2018 Shift] Q#1 (MCQ): "Which of the following statements is true is case of alkyl halides?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is

- (A) Longer and stronger
- (B) Shorter and stronger
- (C) Shorter and weaker
- (D) Longer and weaker **[Correct]**

Correct Answer: D

Solution Analysis: The C-Cl bond in methyl chloride is compared with the C-Cl bond in chlorobenzene. In methyl chloride, CH_3Cl , carbon is sp^3 hybridised and the C-Cl bond is a normal single bond. In chlorobenzene, C_6H_5Cl , the chlorine atom is attached to an sp^2 hybridised carbon of the benzene ring. Also, due to resonance, the lone pair of chlorine overlaps with the benzene ring. This gives the C-Cl bond some partial double bond character. Because of this partial double bond character, the

Benchmark Problem 2: KCET 2018 (Q#1)

Which of the following statements is true is case of alkyl halides?

- (A) They are polar in nature
- (B) They can form hydrogen bonds
- (C) They are highly soluble in water
- (D) They undergo addition reactions

Correct Answer: None

Solution Analysis: Let's analyze each option: Option A: They are polar in nature Alkyl halides have a C-X bond (where X is a halogen such as Cl, Br, or I). The halogen is more electronegative than carbon, which creates a dipole in the bond. Even though the overall molecular polarity can depend on the molecular structure, the presence of the C-X bond makes them polar in nature. Therefore, Option A is true. Option B: They can form hydrogen bonds For a molecule to form hydrogen bonds, it must have a hydrogen atom attached to a highly electronegative atom like O, N, or F.